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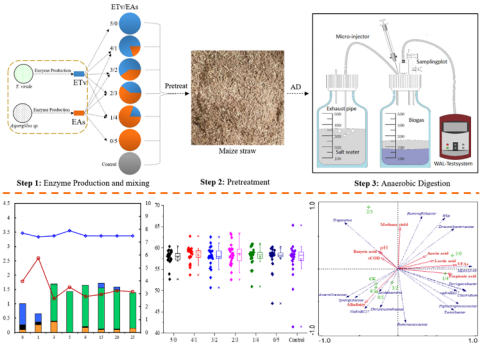
Accelerated biomethane production from lignocellulosic biomass: Pretreated by mixed enzymes secreted by *Trichoderma viride* and *Aspergillus* sp.

Xiaoling Zhao[a,b], Zehui Zheng[a], Yafan Cai[a], Yubin Zhao[a], Yue Zhang[a], Youhui Gao[a], Zongjun Cui[a], Xiaofen Wang[a,*]

[a] College of Agronomy and Biotechnology, China Agricultural University, Beijing 100193, PR China
[b] State Key Joint Laboratory of Environment Simulation and Pollution Control, School of Environment, Tsinghua University, Beijing 100084, PR China

GRAPHICAL ABSTRACT



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ABSTRACT

Biological pretreatment is a promising technology to increase biogas yield. The methane yield and microbial community resulting from anaerobic digestion of maize straw after pretreatment of enzymes [extracted from *Trichoderma viride* (ETv) and *Aspergillus* sp. (EAs)] at different mixing ratios [5/0, 4/1, 3/2, 2/3, 1/4, 0/5] were evaluated. The methane yields from mixed enzymes pretreatment were higher than single enzymes pretreatments of ETv and EAs. The optimal enzymes mixing ratio of ETv and EAs was found to be 2/3, with the cumulative methane yield 512.64 mL/g TS_{added}, which was 31.74% higher than the control. Enzymatic pretreatment promoted an increase in the abundance of bacteria and archaea associated with cellulose decomposition. The majority of bacteria and archaea were assigned to *Bacteroidetes*, *Firmicutes* and *Methanosaeta*.

1. Introduction

Anaerobic digestion (AD) is conducive to the biogas conversion of organic waste. Biogas, methane as its main component, can be purified as bio-natural gas to achieve a complete replacement of natural gas when the methane content is up to 97%. Biogas has the advantage of "zero carbon emissions" in the "carbon cycle", and can reduce the emission of greenhouse gases (GHGs). The annual maize straw (MS) output exceeds 360 million tons in China, which contains abundant potential energy (Zhao et al., 2018). The biogas potential of MS is 0.304 m^3/kg TS, and 9.3 × 10^8 m^3 of biogas can be produced per year if its total solid (TS) content is 85%, with reducing 5.51 × 10^8 kg of CO_2 and replacing 7.0 × 10^8 m^3 of natural gas (Cheng and Zhu, 2012). Straw-to-biogas can not only reduce the environmental pollutions

* Corresponding author.
E-mail address: wxiaofen@cau.edu.cn (X. Wang).

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Fig. 1. VFA, alkalinity/VFA ratio and pH value during AD of MS by pretreated with ETv/EAs at mixing ratio of 5/0, 4/1, 3/2, 2/3, 1/4, 0/5 and the control. ETv, Enzyme secreted by *Trichoderma viride*; EAs, Enzyme secreted by *Aspergillus* sp.

with enzymatic pretreatment significantly improved ($p < 0.05$) the VFA concentration in all biochemical methane potential (BMP) trials (Fig. 1). The initial VFAs is produced by the decomposition of monosaccharide by bacteria attached to the substrate. The 3/2 mixing ratio of the ETv/EAs group had obvious advantages over the others, which meant that the production of monosaccharide in 3/2 group was higher than others. At 0 d, the total VFA concentration reached 3.97 g/L, which was higher than those of the 4/1 (54.19%), 0/5 (98.60%), 5/0 (112.63%), 1/4 (211.29%), and 2/3 (292.49%) groups. In the VFA composition, the concentration of acetic acid, butyric acid, and lactic acid in the 3/2 group was 1.38 g/L, 1.07 g/L, and 0.98 g/L, respectively. The high concentration of acetic acid and butyric acid provided advantageous conditions for AD. On the 1st day, formic acid and acetic acid, which can be easily used by methanogens, were exhausted, only leaving a small part of butyric acid and lactic acid. The acetic acid concentration in the 4/1 group accounted for 47.75% of the total VFA at 0 d, becoming the main component of VFA. With the decrease in the proportion of ETv, the concentration of acetic acid decreased, which may promote the production of acetic acid after pretreatment by ETv. When the ratio of ETv/EAs decreased to 2/3 and 1/4, the VFA concentration decreased, and reduced the VFA composition, indicating that the high proportion of EAs was not conducive to the pretreatment of MS. At 0 d, the concentration of acetic acid in the 5/0 and 0/5 groups increased 130.30% and 238.64%, respectively, compared with the control, and the butyric acid concentration increased by 294.59% and 254.95%, respectively. Acetic acid and butyric acid are small organic acids that can be easily utilized in AD. Butyric acid can be broken down into two molecules of acetic acid, which can be used directly by methanogens (Table 3).

The increase in VFA concentration provided superior initial conditions for AD of MS. After fermentation for 1 d, the VFA concentration of each treatment group changed significantly. Lactic acid can be converted to acetic acid and H_2 by hydrogen-producing acetogen, while its utilization lags behind formic acid and acetic acid. Formic acid, acetic acid, propionic acid, and butyric acid in the 0/5 group was massively consumed, only 0.12 g/L left in total VFA. It was indicated that the easily exploitable organic acids were consumed by methanogens and converted into methane in the AD stage, which could increase the methane yield. In the enzymatic treatment groups, acetic acid and butyric acid were consumed, but propionic acid accumulated, predicting that propionic type fermentation may be carried out in the acidification stage (Kong et al., 2018). From day 2, the concentrations of both butyric and acetic acid sharply decreased, but the propionic acid concentration increased and maintained a high level from days 3 to 25, which was in accordance with the results of studies conducted by Kong et al. (2018). During methane fermentation, the rate at which butyric acid was converted to acetic acid was faster than that at which propionic acid was converted to acetic acid due to the different Gibbs free energy of the conversion (Table 3). In general, the tolerance concentration of methanogens to propionic acid is 1 ~ 2 g/L. Excessive accumulation of propionic acid will cause certain acid inhibition of the AD system. After 1 d of fermentation, the concentration of VFA in each treatment group remained stable, and propionic acid was the primary constituent acid of VFA. With the progress of acid production and microbial metabolism, after the 3rd day, VFA was mainly composed of propionic acid. On the 20th day, the control group had a propionic acid content of 2.42 g/L, and acid inhibition had occurred.

The alkalinity buffers the reaction system. Studies have shown that when the total alkalinity is between 3.00 and 8.50 g/L and the ratio of alkalinity/VFA is more than 2:1, the biogas fermentation will be in a stable state (Lian et al., 1984). In the early fermentation stage (from day 0 to day 5), microorganisms need to adapt to the new environment and establish a dynamic and stable community structure. Alkalinity was increasing and the system was gradually stable. During the fermentation process, the alkalinity value of each group was stable in the range of 3.24 to 5.21 g/L. After 1 d of fermentation, the alkalinity/VFA ratio of 4/1 and 0/5 groups increased sharply to 18.71 and 9.33 in the 1st day treatment, respectively, and then decreased rapidly to 3.17 and 2.40, respectively. The alkalinity/VFA of the 2/3 and 1/4 groups was stable between 2.14 and 5.76, and the fermentation of the two groups was stable. The alkalinity/VFA of the 4/1, 3/2, and 0/5 groups at day 0 was lower than 2:1, indicating that the reaction system required a certain adaptation period.

The decrease or increase of VFA inevitably caused a change in the alkalinity value, and the interaction between the two will maintain the stability of the pH value. The dynamic balance of alkalinity and VFA concentration maintained a permanent pH value between 7.3 ~ 7.9. At the initial fermentation stage, the system was in the acidification stage, and the pH gradually decreased in days 1 ~ 3. The complex organic macromolecules were decomposed into small organic acids, alcohol, CO_2, H_2, etc. by hydrolysis and acidification bacteria, providing necessary nutrients for the subsequent acetate and methane production phases. Subsequently, on the 5th day of fermentation, the pH value increased, indicating that the acidic small molecule substance was consumed, and the concentration of the buffering alkaline substance was increased.

3.3. Methane content and CO_2 content during AD

Besides the enhancement of methane yields, the biological addition of enzymes increased the initial methane production rate in all AD trials (Fig. 2). After 1 d of fermentation, the methane content of treatment groups was higher than 50%, indicating that the methane fermentation started quickly in these groups. The methane content of 5/0 and 0/5 groups was significantly higher than that in control at the 1st d, 26.97% and 13.29% higher, respectively. The methane content of the pretreatment groups was higher than that of control during 0 ~ 5 days, indicating that enzymatic pretreatment can increase initial methane content in the AD of MS. A similar outcome was also mentioned by Nkemka et al. during anaerobic digestion of corn silage and cattail

Table 3
Chemical reaction process of acidification and methanation phase.

Description	Chemical reaction	$\Delta G_0'$ (kJ mol^{-1})*
Conversion of propionic acid to acetic acid, hydrogen	$CH_3CH_2COO^- + 3H_2O \rightarrow CH_3COO^- + HCO_3^- + H^+ + 3H_2$	+76.1
Conversion of butyric acid to acetic acid and hydrogen	$CH_3CH_2CH_2COO^- + 2H_2O \rightarrow 2CH_3COO^- + H^+ + 2H_2$	+48.1
Conversion of ethanol to acetic acid and hydrogen	$CH_3CH_2OH + H_2O \rightarrow CH_3COO^- + H^+ + 2H_2$	+9.6
Conversion of lactic acid to acetic acid and hydrogen	$CH_3CHOHCOO^- + 2H_2O \rightarrow CH_3COO^- + H^+ + HCO_3^- + 2H_2$	−4.2
Hydrogen methanogenesis	$HCO_3^- + 4H_2 + H^+ \rightarrow CH_4 + 3H_2O$	−135.6
Acetate methanogenesis	$CH_3COO^- + H_2O \rightarrow CH_4 + HCO_3^-$	−31.0

*T = 298 K, pH = 7.0, $p = 1.0 \times 10^5$ Pa, product concentration is 1 mol L^{-1}.

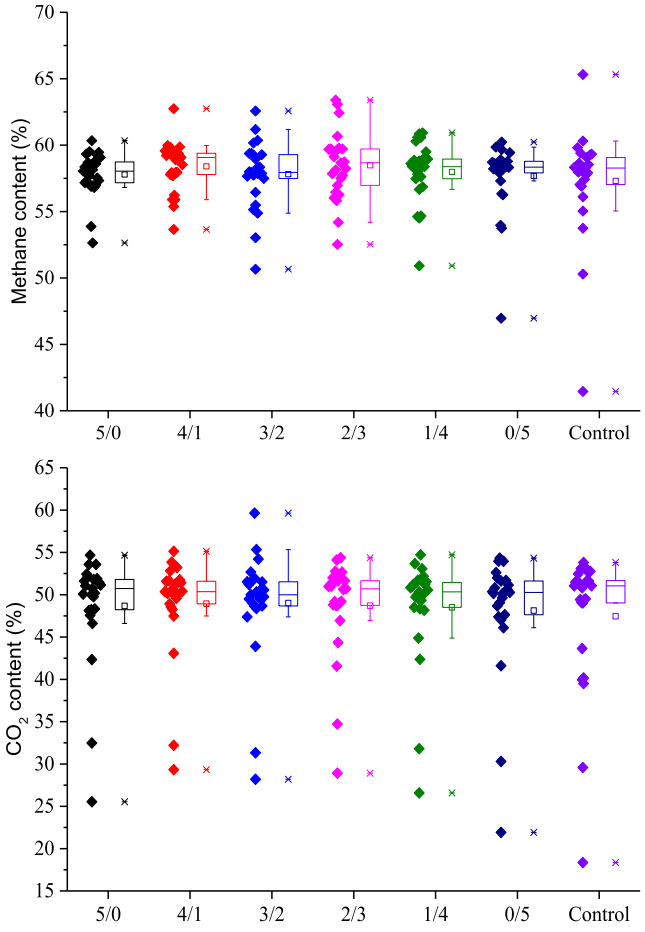

Fig. 2. Methane content and CO_2 content during AD of MS by pretreated with ETv/EAs at mixing ratio of 5/0, 4/1, 3/2, 2/3, 1/4, 0/5 and the control. ETv, Enzyme secreted by *Trichoderma viride*; EAs, Enzyme secreted by *Aspergillus* sp.

(Nkemka et al., 2015). Methane content in the 2/3 group was always higher than 60% at days 2 ~ 4, predicting that the methanogenesis was active in methane metabolism, and having high utilization of nutrients. Methane content of 4/1, 3/2, and 1/4 group also reached the maximum at day 3, which were 62.74%, 62.57%, and 60.85%, respectively. However, methane content of the control group was only 56.91% on day 3, then decreased to 50.30% at day 4, having poor methane production ability, indicating that the enzymatic pretreatment accelerated the start-up time of methane fermentation. Methane content during days 10 ~ 20 showed the same trend, which was between 53.66% and 60.37%, having a stable AD process. In other words, enzymes contributed to shortening the microbial lag-phase of anaerobic biodegradation, which can be worth noting to reduce the retention time of biogas reactors.

The median value of CO_2 content in the enzymatic pretreatment groups was lower than that of the control group, and the content distribution was concentrated. The average value was higher than that of the control group, indicating that the methane fermentation process was more stable after enzymatic pretreatment.

3.4. Methane yield

According to BMP results, enzymes with different proportions boosted the methane yield by 6.08–31.74% in all examined trials in this particular study. Daily methane yield of enzymatic treatment groups was significantly higher than that of the control group on day 1, because the pretreatment process promoted the hydrolysis of MS. At the beginning of AD, the VFA concentration increased, and nutrients that can be directly used by microorganisms also increased, so the activities of methane metabolism were more active than control. The daily methane yield of 2/3 group was the largest (62.11 mL/g TS_{added}) at day 1, which was 42.39%, 24.12%, 47.63%, 46.87%, 51.30%, and 57.00% higher than 5/0, 4/1, 3/2, 1/4, 0/5 group, and the control, respectively (Fig. 3). At day 2, the daily methane yield of treatment groups decreased to 24.13 ~ 28.51 mL/g TS_{added}, then rose to 44.10 ~ 47.45 mL/g TS_{added} at day 3.

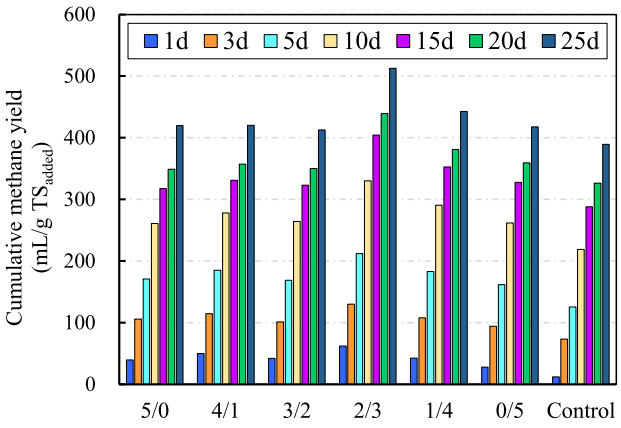

Fig. 3. Cumulative methane yield on different days during AD of MS by pretreated with ETv/EAs at mixing ratio of 5/0, 4/1, 3/2, 2/3, 1/4, 0/5 and the control. ETv, Enzyme secreted by *Trichoderma viride*; EAs, Enzyme secreted by *Aspergillus* sp.

The cumulative methane yield in the control group was always lower than that in the enzyme-treated groups (Fig. 3), which indicated that the enzymatic pretreatment produced effective MS hydrolysis, and that the hydrolysate could be utilized by the microorganisms in the fermentation system to be converted to methane. The 5/0 group was slightly better than 0/5 group during days 0 ~ 12, but cumulative methane yield was 419.63 and 417.81 mL/g TS_{added} at day 25, respectively, which was no significant difference. Cumulative methane yield in the 2/3 group reached 512.64 mL/g TS_{added}, 31.74% higher than that in the control group. In the current study, the medium containing maize straw powder and wheat bran was used to produce complex enzymes, which reduced the production cost of enzymes. The result was better than only using a single enzyme to pretreat maize straw from Schroyen et al., who used laccase to pretreatment and obtained a 17% increase of methane production (Schroyen et al., 2014). It can be seen that enzymatic treatment had enhanced methane production ability, and mixed enzymes were better than single enzymes. Compared to pretreatment by the microorganism, mixed enzymes also have an obvious advantage in the actual effect. Because microorganisms consumed large amounts of soluble organic matter, which limited the applied effect (Yuan et al., 2014; Zhao et al., 2018).

3.5. Kinetic analysis of methane production

Modified Gompertz model describes the relationship between cumulative methane yield and time (Zwietering et al., 1990), including the lag period, the maximum methane production rate, and methane-producing potential (Lay et al., 1998; Zwietering et al., 1990). The correlation coefficient (R^2) of each treatment was between 98.09% and 98.92% (Table 4). There was no significant difference between the predicted value and the actual value of methane yield, indicating that the Modified Gompertz model can reflect the methane production of MS during AD (Zhao et al., 2018).

The λ value reflects the reaction rate of the substrates in the AD system, which is indirectly reflected in the impact strength of the material on microorganisms. A low λ value indicates that microorganisms

Table 4
Methane production kinetic analysis.

Treatments	Predicted methane yield (mL/g TS_{added}) P_{max}	Obtained methane yield (mL/g TS_{added}) P	Rmax (mL/g TS_{added}·d) R_{max}	Period of fermentation delay λ (d)	Correlation coefficient R^2
5/0	396.65 ± 9.73 ^{cd}	419.63 ± 13.97^{bc}	23.89 ± 1.48	−1.44	0.9809
4/1	393.89 ± 7.92 ^{cd}	420.15 ± 8.22^{bc}	26.30 ± 1.62	−1.46	0.9811
3/2	393.35 ± 8.07 ^{cd}	412.78 ± 16.02^{c}	24.98 ± 1.38	−1.16	0.9847
2/3	493.01 ± 9.31^{a}	512.64 ± 21.53^{a}	30.67 ± 1.55	−1.40	0.9872
1/4	420.60 ± 7.33^{b}	442.74 ± 8.17^{b}	28.62 ± 1.46	−0.85	0.9870
0/5	404.34 ± 7.48^{c}	417.81 ± 7.95^{c}	25.69 ± 1.20	−0.61	0.9892
Control	385.82 ± 9.97^{d}	389.13 ± 7.25^{d}	21.83 ± 1.08	−0.15	0.9877

Table 5
Alpha diversity estimators of bacteria and archaea.

Sample	Bacteria							Archaea						
	Seq_num	sobs	shannon	simpson	ace	chao	coverage	Seq_num	sobs	shannon	simpson	ace	chao	coverage
5/0	30,283	851	4.81	0.03	946.28	947.23	0.9948	30,313	56	2.15	0.18	59.29	59.0	0.9998
4/1	41,751	879	5.11	0.02	921.78	922.56	0.9977	44,549	51	1.83	0.26	57.08	54.5	0.9998
3/2	30,275	841	5.04	0.02	921.43	917.71	0.9954	35,915	51	1.85	0.26	52.95	53.0	0.9999
2/3	41,234	855	4.84	0.03	928.20	944.90	0.9966	34,212	53	2.07	0.18	61.66	60.2	0.9997
1/4	42,374	856	4.79	0.03	910.19	921.94	0.9973	39,500	57	2.00	0.21	61.74	64.0	0.9998
0/5	36,004	849	4.98	0.02	924.25	959.08	0.9962	33,687	53	1.91	0.23	57.77	56.0	0.9998
Control	30,277	847	5.12	0.02	913.70	908.69	0.9960	41,191	52	1.90	0.23	54.78	54.5	0.9999
Sludge	39,552	775	4.34	0.05	867.70	880.76	0.9959	37,890	49	1.95	0.22	50.95	50.5	0.9999

can adapt quickly to the new environment with no significant delay, so the fermentation system can rapidly generate methane. Conversely, a high λ value indicates that microorganisms need some time to adapt to the new environment and that methane production is delayed (Zhao et al., 2018). In this study, λ values of each treatment group were all less than 1 d, indicating that microorganisms could quickly adapt to the external environment, and methane could be rapidly produced on the first day under anaerobic conditions. Λ values of enzymatic pretreatment groups were smaller than that of the control group (-0.15 d), indicating that enzymatic pretreatment provided a more suitable environment for AD, which could be seen from the initial concentration of VFA in Fig. 1. Enzymatic pretreatment improved the initial VFA in the anaerobic system, and small molecule organic acids were the substrate for methanogens in AD. This favorable condition promoted the rapid response of methanogens to the environment. It can be seen from Fig. 3 that the daily methane yield of enzymatic treatment groups was significantly higher than that of the control group at the initial stage of AD. Enzymatic pretreatment created superior conditions for AD of MS and provided more nutrients for anaerobic microorganisms. The λ values of enzymatic treatment groups were all less than 0 d, reflecting that the appropriate pretreatment conditions played a decisive role in the efficient operation of AD.

The modified Gompertz model predicts the maximum methane production rate and reflects the methane production performance. The 2/3 group had the largest R_{max} (36.92 mL/g TS_{added}), which showed that methane production performance was optimal under the treatment conditions. The control group had the lowest R_{max} (23.36 mL/g TS_{added}), which may be related to the low initial VFA concentration. Because in the early stage of fermentation, methanogens can only use the existing small molecular acid in the environment to synthesize methane. As the fermentation process progresses, the decomposing bacteria play a decomposing function. When the macromolecular organic matter is decomposed into small molecules of acid, H_2, and CO_2, the methanogen will undergo subsequent growth and metabolism.

3.6. Effect of enzymatic pretreatment on the microbial community in AD

In agricultural production, *Trichoderma* sp. are often used as biological control agents to control crop diseases by inhibiting pathogenic microorganisms (Moya et al., 2020). Zhou found that *Aspergillus niger* have a certain inhibitory effect on marine bacteria (Zhou et al., 2014). Anaerobic digestion is a microbial process involving hydrolytic bacteria and methanogenic archaea. The mixed enzymes used in the study were extracted from *Trichoderma viride* and *Aspergillus* sp. by solid-state fermentation. Whether the extract contains substances that inhibit the growth of microorganisms is essential for AD. Therefore, 24-hour samples were obtained from the AD system and determined by Illumina Miseq high-throughput sequencing technology to evaluate the effect of the enzymes on the microbial system. The primers pair 338F/806R and 524F/958R were used to amplify the V3 ~ V4 regions of bacteria and archaea. The average high-quality sequence reads of bacteria and archaea were 36468 and 37157, and the sequencing coverage was 99.48% and 99.97%, respectively. These data were sufficient to ensure the accuracy of the rarefaction analysis, and all rarefaction curves reached a plateau. Table 5 showed alpha diversity indices of the bacteria and archaea, characterized by sobs, shannon, simpson, ace, and chao indices. Sobs, ace, and Chao describe the changes in the community richness after enzymatic pretreatment. Shannon and Simpson were used to reflect the diversity of the microbial communities in the AD system. Sobs value was the actual observation of microbial community richness, ie, the total number of microorganisms in the samples. The bacteria sobs value in the inoculated sludge was low, then increased after the addition of MS. Mixed enzymes improved the richness and diversity of bacterial communities, and provided a basis for enhancing the stability of the AD system.

The archaea's sobs value was low (51 ~ 57), which was equivalent to the content of archaea in the sludge, which was much lower than that of bacteria. The possible reason was that methanogens had a longer generation time, and the samples were measured at 24 h fermentation. Methanogens did not have enough time to grow and multiply. Nevertheless, selecting this time to determine methanogens was more necessary to show the enzyme's impact on the microorganisms in the fermentation system. If microorganisms could not be stably present in the reaction system, even if the action of the enzyme accelerated the cellulose decomposition, it also not resulted in high methane yield. The activity of the enzyme was recognized if it did not bring harmful substances to the fermenting microorganism. Compared with the sludge and control groups, the richness and diversity of the enzymatic

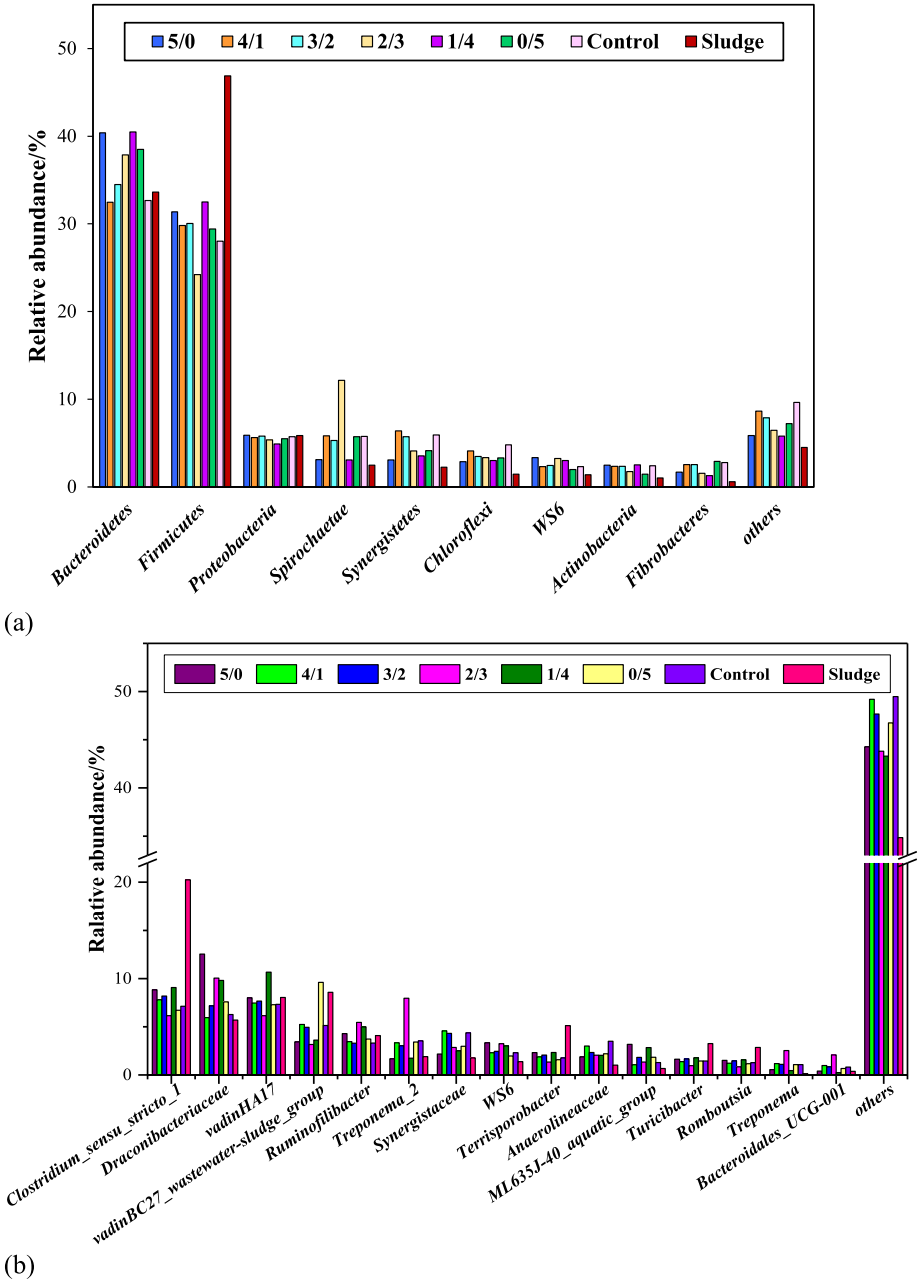

Fig. 4. Bacterial community diversity in (a) phylum level and (b) genus level (abundance > 2%).

pretreatment did not cause large oscillations, indicating that the use of enzymes did not inhibit the methanogens. It ensured that smaller molecular substances would be available by methanogens under the enhanced bacterial activity, which would increase the methane production capacity of MS.

3.6.1. Bacterial community

At the phylum level (Fig. 4a), *Bacteroidetes*, *Firmicutes*, *Proteobacteria*, *Synergistetes*, *Spirochaetae*, *Choloflexi* were dominant in the AD system. The abundance of *Bacteroidetes* and *Firmicutes* was more than 60% of the total bacteria. *Bacteroidetes* were the main bacteria in the anaerobic fermentation process. *Firmicutes* were widely found in anaerobic reactors and rumen, which were significant microorganisms for acidification and hydrolysis. Both phyla have the function of releasing extracellular enzymes to enhance the decomposition and hydrolysis of lipids, proteins, cellulose, and hemicellulose (Kong et al., 2018). *Bacteroidetes* in the enzymatic treatment groups increased, that is, the abundance of *Bacteroidetes* in the 5/0, 1/4, and 0/5 group increased to 38.24%~40.30%, which was 18.06%~24.42% higher than that of the control group. The abundance of *Firmicutes* in the inoculated sludge was the most (47.10%), while decreased in the control and pretreatment groups. The *Firmicutes* abundance decreased the most in the 2/3 group, dropping to 24.42%. According to Table 4, methane yield of 2/3 group was 512.64 mL/g TS_{added}, 31.74% higher than the control group; interestingly, in these treatment groups, methane yield increased with the decrease of *Firmicutes* abundance. *Firmicutes* abundance affected methane production, and when the abundance of *Firmicutes* was low, methane yield increased. Meanwhile, the higher the total abundance of *Bacteroidetes* and *Firmicutes*, the lower the methane yield, such as the 3/2 group; conversely, the higher the methane yield, such as the 2/3 group. It can be seen that the bacterial community diversity had a more significant impact on methane production.

Proteobacteria, *Spirochaetae*, *Synergistetes*, *Choloflexi*, *WS6*, *Actinobacteria*, and *Fibrobacteres* accounted for lower abundance in anaerobic reactors, with a total proportion of 16.07%~31.48%. Such microorganisms were a relatively small part in the AD system, but also played an important role. *Proteobacteria*, *Spirochaetae*, and *WS6* were common in anaerobic reactors, which played a role in the decomposition of organic macromolecules. But *Proteobacteria* and *WS6* did not change much among the pretreatment groups and control group and inoculated sludge. *Spirochaetae* was found in the termites rumen, which can use a variety of carbohydrates, including cellulose and hemicellulose, and synthesize acetic acid with H_2 and CO_2 for methanogens (Leadbetter et al., 1999). This phylum of bacteria changed in different treatment groups, with the abundance increased to 12.16% in the 2/3 treatment group, indicating that the hydrolysis and acid production process was enhanced after the enzyme pretreatment with the 2/3 mixing ratio.

At the genus level (Fig. 4b), *Clostridium*, *VadinHA17*, *Draconibacteriaceae*, *vadinBC27*, and *Ruminofilibacter* were dominant. In the 2/3 group, with the highest methane yield, the abundances of *Ruminofilibacter*, *Treponema 2*, *Treponema,* and *Bacteroidales_UCG-001* were higher than those of other treatment groups, with abundance values of 5.40%, 7.90%, 2.50%, and 2.06%, respectively. These four types of bacteria all had the ability to decompose cellulose, which may be the reason for the high methane yield in the 2/3 group. It also reflected that the pretreatment of MS with the 2/3 mixing ratio of ETv/EAs conducive to the use of cellulose by bacteria in AD.

3.6.2. Archaeal community

At the genus level (Fig. 5), the dominant archaea included methanogenic archaea and non-methanogenic bacteria. The methanogenic archaea contained five genera, *Methanosaeta*, *Methanosarcina*, *Methanebacterium*, *Methanomassiciococcus*, and *Methanocorpusculum*, with the total abundance accounts for 37.57%~67.57% in archaea. *Methanosaeta* was the dominant methanogen in the anaerobic system, accounting for 14.18%~44.59%, but the abundance among different treatment groups was significantly different. *Methanosaeta*, which are always found in a medium temperature environment, have an acetic acid metabolism and specifically produce methane from acetic acid (Großkopf et al., 1998). Followed by *Methanosarcina*, the abundance of *Methanosarcina* between the treatment groups did not change much and was in a stable state. The archaea are widely found in methane

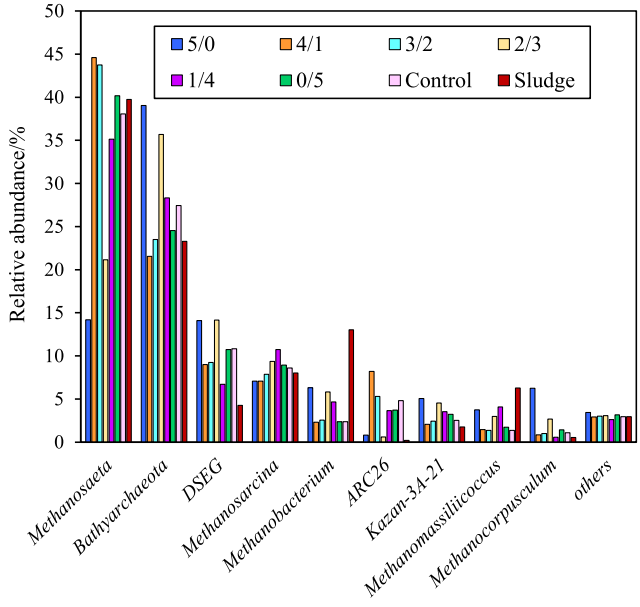

Fig. 5. Archaeal community diversity in genus level (abundance > 2%).

fermentation reactors. They are usually the dominant bacterium for methane production, which are acetic nutritional microorganisms, mainly found in systems with high concentrations of VFA (especially acetic acid), and their abundance was affected by the concentration of acetic acid (Zhao et al., 2017b). The abundance of *Methanebacterium* and *Methanomassiciococcus* in the pretreatment group was lower than that of sludge (both *Methanebacterium* and *Methanomassiciococcus* accounted for 19.29% of the total abundance). *Methanebacterium* can produce methane using H_2 and CO_2 (Krivushin et al., 2010). *Methanomassiliicoccus* had a unique metabolic property and was a methyl-type metabolism (Iino et al., 2013). However, unlike traditional obligate methyl-type methanogens, this type of archaea lacked the complete pathway of reducing CO_2 to methyl-CoM (Lang et al., 2014). So, these archaea required additional H_2 to grow. Genomic analysis revealed that *Methanomassiliicoccus* contains genes that reduce methanol, methylamine, dimethylamine, and other substrates by H_2. Therefore, *Methanomassiliicoccus* is neither an atypical methylotrophic methanogen nor an atypical hydrogen methanogen, but a mixed type of both (Lang et al., 2014; Brugère et al., 2014). *Bathyarchaeota* had a variety of metabolic modes, which can degrade macromolecules such as aromatic compounds, chitin, cellulose, and protein, and produce small molecular compounds such as acetic acid by fermentation. The energy was also obtained by autotrophic acetogenic pathways using CO_2 and H_2. Acetic acid produced by *Bathyarchaeota* is a major electronic carrier that can be utilized by methanogens and heterotrophic bacteria, and is a core driver of the deep carbon cycle and ecosystem (He et al., 2016). It speculated that *Deep sea Euryarchaeotic Group-DSEG* had a positive effect on cellulose decomposition. Therefore, in the anaerobic reactor with straw as a substrate, the large number of such archaea will facilitate the decomposition of cellulose into small molecules, providing substrates for methanogens (Zhao et al., 2018). Although the methanogen abundance in the pretreatment groups was lower than that of sludge, the number of archaea associated with cellulose degradation increased, which was beneficial in early AD.

3.7. Relationship between microorganisms and environmental factors

Redundancy analysis (RDA) was used to visually analyze the relationship between environmental factors and microorganisms (Fig. 6). The 4/1, 3/2, 0/5, and the control treatment groups were concentrated in the third quadrant, and there was no significant difference in methane content among 4/1, 3/2, and 0/5 group. The 2/3, 5/0, and 1/4 groups, which obtained higher methane yield, were more dispersed, farther to the main component axis. Methane yield was positively correlated with acetic acid, butyric acid, and sCOD, indicating that methane was mostly converted from acetic acid, butyric acid, and sCOD. Propionic acid formed the largest proportion of VFA and was highly correlated with VFA. According to Fig. 1, propionic acid was the main constituent acid in MS single digestion, while lactic acid, acetic acid, and butyric acid accounted for a small proportion in VFA. Alkalinity and VFA together maintained the acid-base balance in the AD system, in which acetic acid and butyric acid made a greater contribution in maintaining acid-base balance. The bacteria were mostly related to the degradation of cellulose or the conversion to small organic acids. *Ruminofilibacter*, *Treponema*, *WS6*, *vadinHA17*, *Ruminococcaceae*, *Clostridium*, *VadinBC27*, etc. were abundant in the AD system, and playing an important role in acid conversion. It is obvious that bacteria in the treatment groups with high methane yield, such as 2/3 and 1/4 group, was richer and more relevant. It can be seen that pretreatment conditions affected the microbial community structure, and then the microbes changed the fermentative methane production capacity by changing relevant indicators in the AD environment.

4. Conclusions

Enzymatic pretreatment changed the initial VFAs concentration and

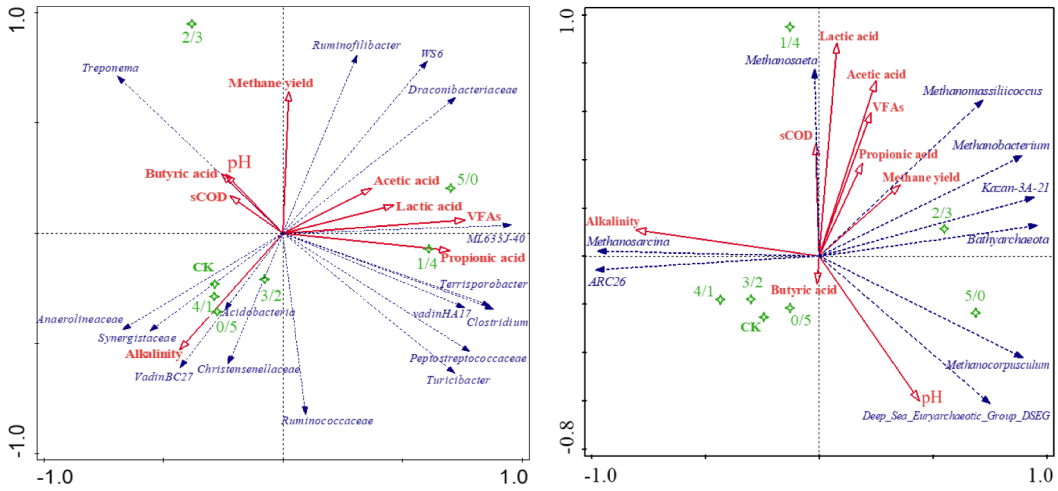

Fig. 6. Redundancy analysis between microbial and environment (bacteria and archaea).

composition, and the conversion rate of MS to biomethane was higher after enzymatic pretreatment. The predominant VFA detected in all groups was propionic acid. Mixed enzymes performed better than single enzymes, and a 2/3 mixing ratio of ETv/EAs showed the optimal performance, achieving the largest cumulative methane yield (512.64 mL/g TS_{added}). Simultaneously, enzymatic pretreatment stimulated the abundance and diversity of bacterial and archaeal communities associated with cellulose degradation, providing more nutrients for methanogens. *Bacteroides* and *Firmicutes* were the dominant bacteria, and *Methanosaeta* was the dominant archaea in AD.

CRediT authorship contribution statement

Xiaoling Zhao: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Zehui Zheng:** Conceptualization, Writing - review & editing. **Yafan Cai:** Project administration, Writing - review & editing. **Yubin Zhao:** Project administration. **Yue Zhang:** Project administration. **Youhui Gao:** Project administration. **Zongjun Cui:** Conceptualization, Methodology, Resources, Funding acquisition. **Xiaofen Wang:** Conceptualization, Methodology, Resources, Writing - review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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